SNN Atan α Convergence Analysis vs CNN Baseline (MNIST)

ByzFL SNN Benchmark

July 2, 2026

1 Overview

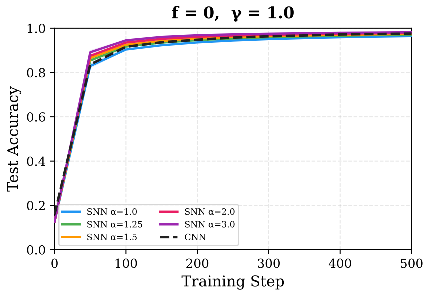
This document presents training convergence plots comparing Spiking Neural Network (SNN) models with different ArcTangent surrogate gradient scaling factors $\alpha \in \{1.0, 1.25, 1.5, 2.0, 3.0\}$ and the CNN baseline.

Experimental settings:

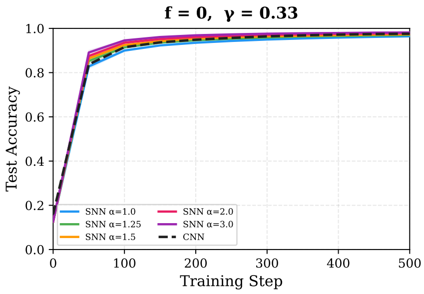
- **Dataset:** MNIST
- **Aggregator:** GeometricMedian (with NNM + ARC pre-aggregation)
- **Attack:** Optimal A Little Is Enough
- **Honest clients:** $n_h = 10$, Byzantine: $f \in \{0, 3, 5\}$
- **Data heterogeneity:** $\gamma \in \{1.0, 0.33, 0.0\}$ (IID $\rightarrow$ extreme non-IID)
- **Seeds:** 5 seeds (mean $\pm$ std shading)

2 Test Accuracy over Training Steps

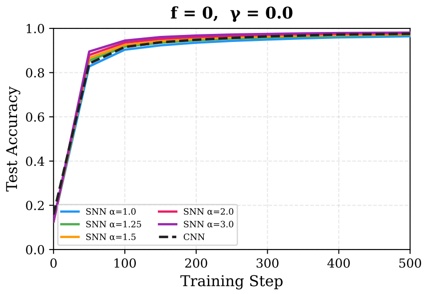
Each panel shows test accuracy as a function of training step for SNN with different α values (solid colored lines) and the CNN baseline (dashed black).



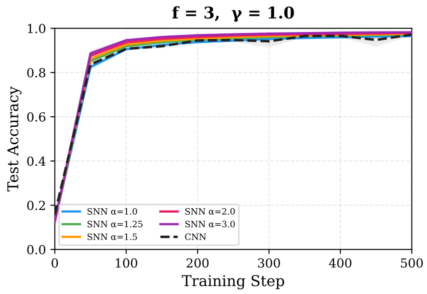
(a) $f = 0$, $\gamma = 1.0$



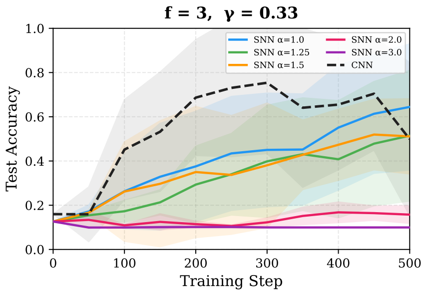
(b) $f = 0$, $\gamma = 0.33$



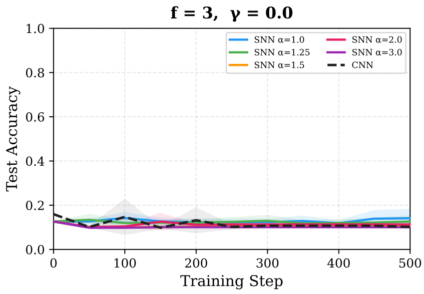
(c) $f = 0$, $\gamma = 0.0$



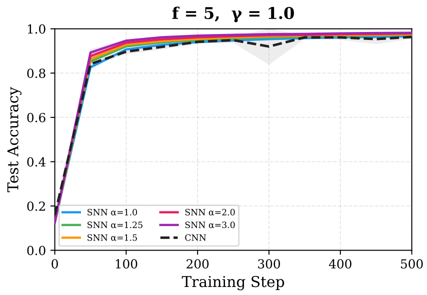
(d) $f = 3$, $\gamma = 1.0$



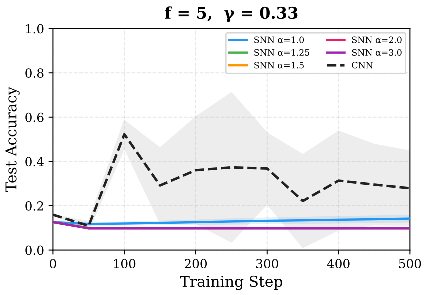
(e) $f = 3$, $\gamma = 0.33$



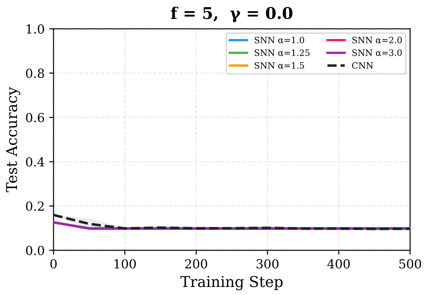
(f) $f = 3$, $\gamma = 0.0$



(g) $f = 5$, $\gamma = 1.0$



(h) $f = 5$, $\gamma = 0.33$

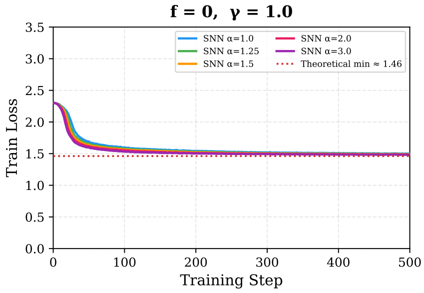


(i) $f = 5$, $\gamma = 0.0$

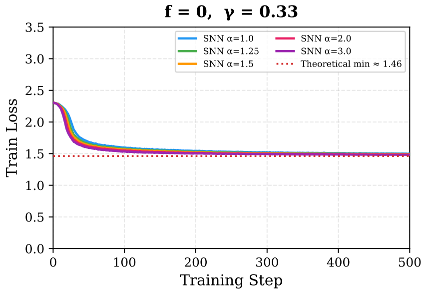
Figure 1: Test Accuracy convergence: SNN ($\alpha \in \{1.0, 1.25, 1.5, 2.0, 3.0\}$) vs CNN baseline. Rows: increasing Byzantine fraction (f). Columns: decreasing data IID-ness (γ).

3 Train Loss over Training Steps (SNN Only)

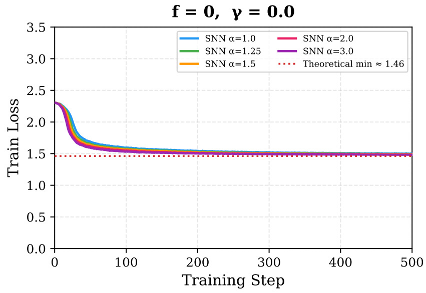
Train loss for SNN models only. CNN loss is omitted because the loss functions are not comparable (SNN uses rate-coded cross-entropy, CNN uses NLL loss).



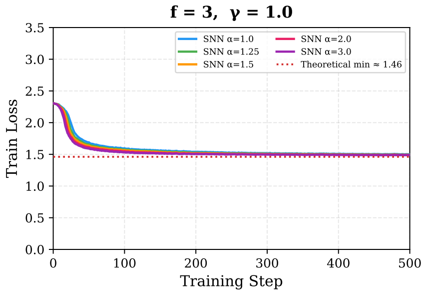
(a) $f = 0$, $\gamma = 1.0$



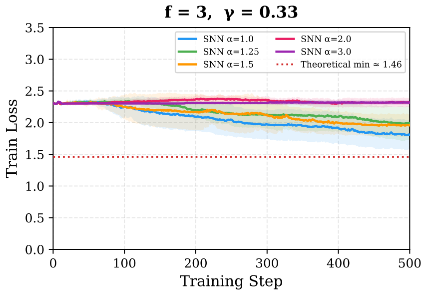
(b) $f = 0$, $\gamma = 0.33$



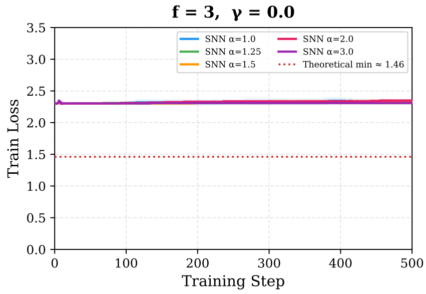
(c) $f = 0$, $\gamma = 0.0$



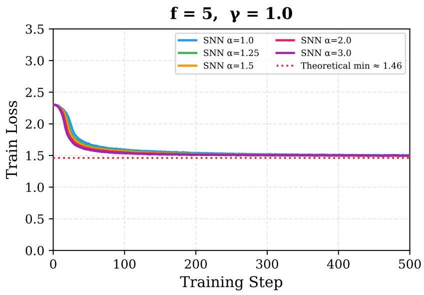
(d) $f = 3$, $\gamma = 1.0$



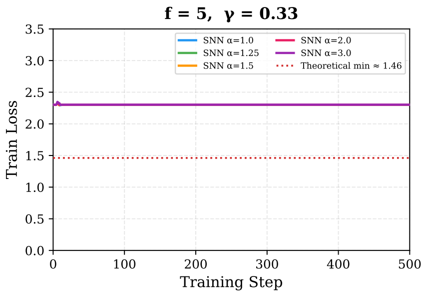
(e) $f = 3$, $\gamma = 0.33$



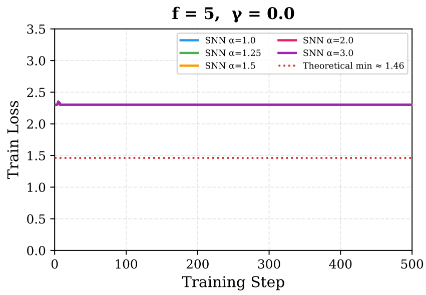
(f) $f = 3$, $\gamma = 0.0$



(g) $f = 5$, $\gamma = 1.0$



(h) $f = 5$, $\gamma = 0.33$

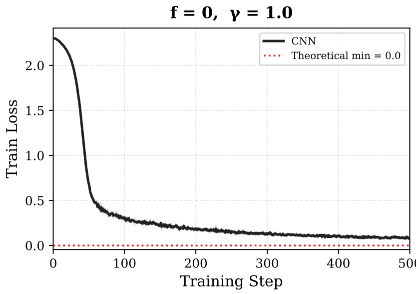


(i) $f = 5$, $\gamma = 0.0$

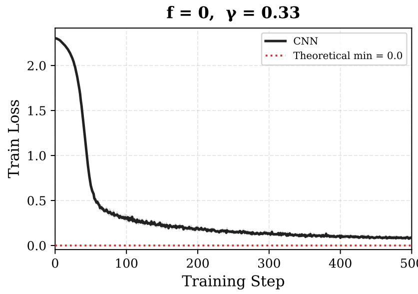
Figure 2: Train Loss convergence for SNN only ($\alpha \in \{1.0, 1.25, 1.5, 2.0, 3.0\}$). CNN is excluded because loss functions are not directly comparable.

4 Train Loss over Training Steps (CNN Only)

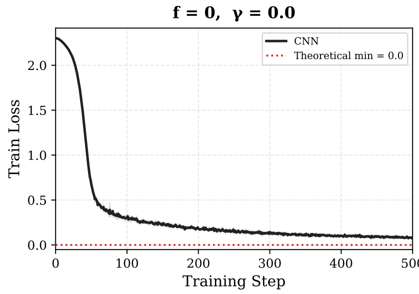
Train loss for the CNN baseline. Shown separately from SNN because the loss functions differ (CNN uses NLL loss, SNN uses rate-coded cross-entropy).



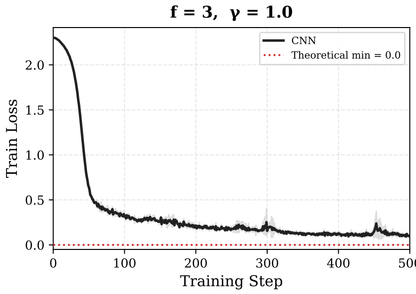
(a) $f = 0$, $\gamma = 1.0$



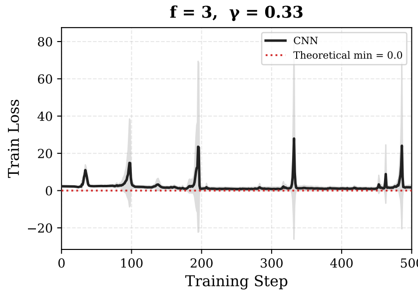
(b) $f = 0$, $\gamma = 0.33$



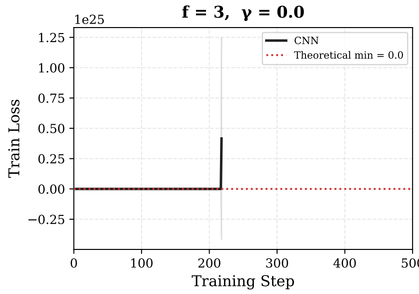
(c) $f = 0$, $\gamma = 0.0$



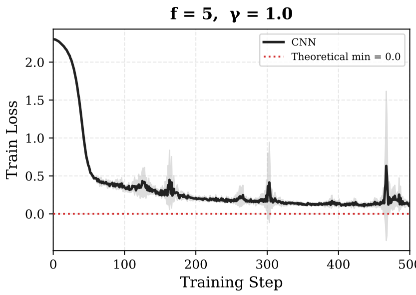
(d) $f = 3$, $\gamma = 1.0$



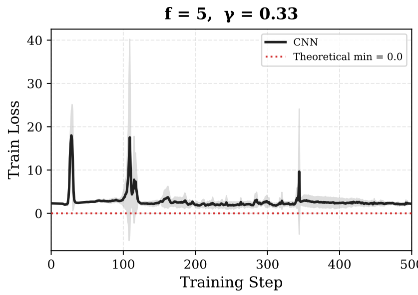
(e) $f = 3$, $\gamma = 0.33$



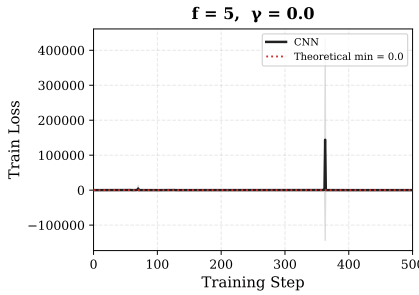
(f) $f = 3$, $\gamma = 0.0$



(g) $f = 5$, $\gamma = 1.0$



(h) $f = 5$, $\gamma = 0.33$



(i) $f = 5$, $\gamma = 0.0$

Figure 3: Train Loss convergence for CNN baseline only (NLL Loss).

5 Key Observations

- **Benign settings** ($f = 0$, any γ): All α values converge similarly; higher α is slightly faster initially.
- **Moderate adversarial** ($f = 3$, $\gamma = 0.33$): $\alpha = 1.0$ shows the most robustness, climbing to ∼65% while $\alpha \geq 2.0$ collapse to random. CNN achieves higher but unstable accuracy.
- **IID under attack** ($f = 3$ or $f = 5$, $\gamma = 1.0$): Surprisingly, higher α converges faster even under attack when data is IID.
- **Extreme adversarial** ($f = 5$, $\gamma = 0.0$): All models collapse to random accuracy.
- **Train loss stability**: SNN train loss is much smoother than CNN under Byzantine attack, suggesting intrinsic gradient stability of spiking networks.